Gauge Specifications.

This document outlines the specifications for headspace gauges for the following calibers:

CALIBER	"F" (GO- MINIMUM)) +.0002 (inches)	"F" (NO-GO MAXIMUM)) -.0002 (inches)	DIAMETER "M" +0.002
6.5 Creedmoor	1.541	1.551	.400
7.62 x 39	1.252	1.262	.3662
7.62x51 (NATO)	1.6300	1.6400	.400
300 AAC Blackout	1.0789	1.0889	.351
300 Norma Magnum	2.036	2.046	.455
300 Winchester Magnum	.220	.227	.533
338 Norma Magnum	2.0101	2.0201	.475
5.56mm (NATO, ARMY)	1.4646	1.4730	.330
277 Fury	1.6800	1.6870	.400

see SAAMI reference drawing at the end of this document

** SAAMI reference for 300 Winchester magnum at the end of this document "Belt Breech".

1. Headspace Dimensions:

- **Go Gauge (Minimum Headspace):** SAAMI or NATO MINIMUM headspace dimension +0.0002" tolerance. – as applicable (See Caliber)
- **No-Go Gauge (Maximum Headspace):** SAAMI or NATO MAXIMUM headspace dimension -0.0002" tolerance. – as applicable (See Caliber)
- **Diameter Requirements:** Gauges must conform to SAAMI or NATO diameter specifications at both the MINIMUM (Go Gauge) and MAXIMUM (No-Go Gauge) headspace distances.

2. Material and Hardness:

- **Material:** High-alloy tool steel AISI O1 or O2.
- **Heat Treatment:** SAAMI Heat treated to a Rockwell Hardness of 58-60 HRC.

3. Geometric Tolerances:

- **Perpendicularity:** The critical surfaces of the gauge (e.g., datum surfaces to gauge faces) must be perpendicular within 0.0002" Total Indicated Reading (TIR).

4. Surface Finish:

- **Surfaces:** All bearing surfaces (i.e., surfaces that contact the chamber) must have a surface finish of 16 microinches Ra (Roughness Average) or better. All other surfaces must have at minimum surface finish of 32 microinches Ra or better.

5. Labeling:

- Each gauge must be permanently and clearly labeled with the following information:
    - "SAAMI MINIMUM (GO)" **OR** "NATO MINIMUM (GO)" - as applicable
    - "SAAMI MAXIMUM (NO-GO)" **OR** "NATO MAXIMUM (NO-GO)" - as applicable
    - Nominal headspace dimension (e.g., "1.6300")
    - Caliber (e.g., "5.56x45mm")
    - Manufacturer's identification (e.g., serial number or lot number)

6. Certification and Calibration:

- The vendor must provide third-party certification/calibration for the following critical dimensions, verified to tolerances per applicable SAAMI or NATO specifications:
    - Go Gauge (Minimum Headspace dimension) + .0002”
    - No-Go Gauge (Maximum Headspace dimension) -.0002”
    - Perpendicularity
    - Diameter
- The certification/calibration report must include:
    - Gauge identification (serial/lot number)
    - Date of calibration
    - Calibration laboratory name and accreditation information (e.g., ISO 17025)
    - List of instruments used for calibration (including calibration dates)
    - Measurement data for each critical dimension
    - Statement of conformity to SAAMI or NATO specifications
    - Signature and title of the certifying technician.

7. CAD Drawings:

- The vendor must provide a complete and accurate CAD drawing for each gauge, including:
    - All critical dimensions and tolerances
    - Material specifications
    - Heat treatment specifications
    - Surface finish requirements
    - Datum surfaces
    - Revision level

8. Packaging and Storage:

- **Individual Packaging Required:** Each gauge *must* be individually packaged in a labeled, durable storage container to protect it from damage during shipping and storage and to facilitate proper storage of the gauges when not in use.

- **Container Requirements:** The storage container must be appropriate for the gauge size and weight and made of a material that will not damage the gauge (e.g., a rigid plastic case with foam padding).
- **Labeling:** The label on the storage container must match the information on the gauge.